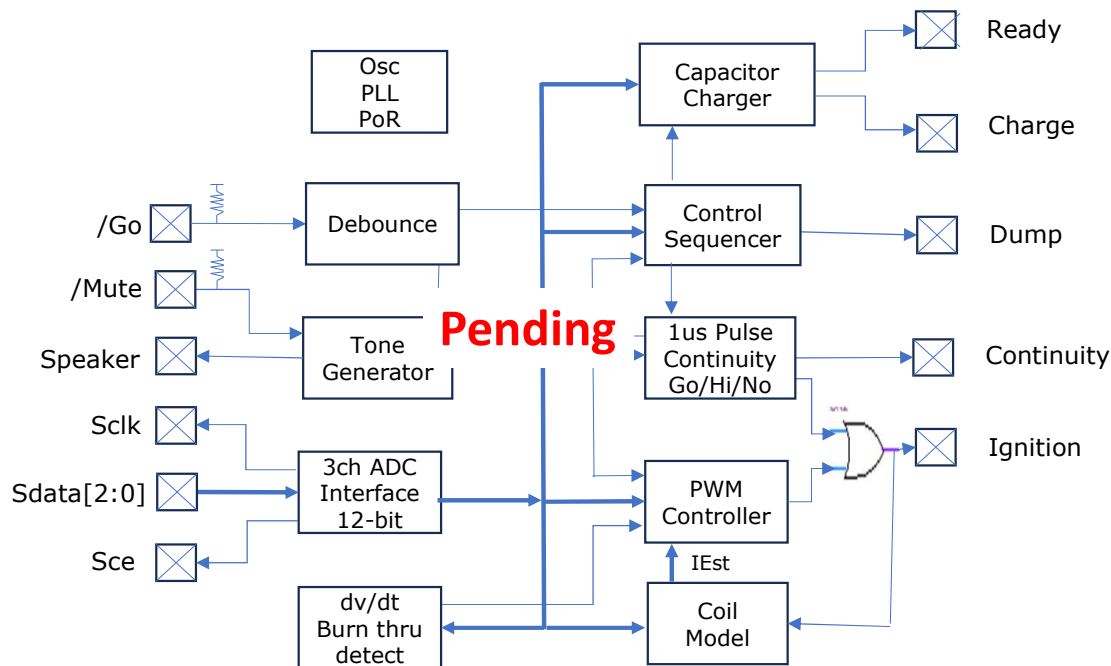


XS-PCC-01 Power Control Chip Block Diagram



Purpose

- Ultra-compact controllers delivering stable micro-grid formation, energy balancing, and dump-load regulation.

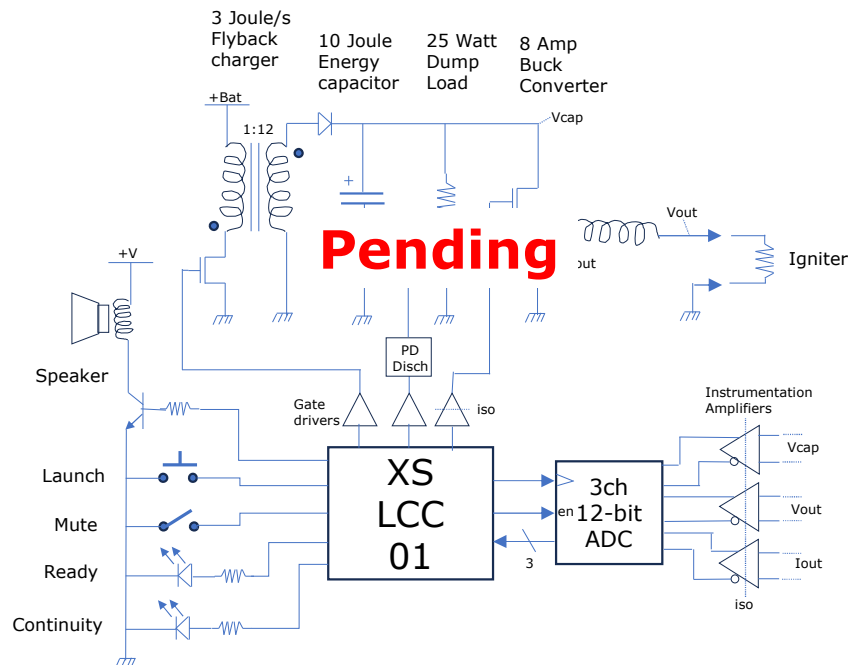
Features

- 48 MHz fully synchronous digital core
- Dual 3 MHz differential ADC inputs with isolation-friendly front-end
- Integrated AC-loop controller with CORDIC-based sine synthesis
- Integrated DC-loop controller with energy-balance regulation
- Deterministic timing — no firmware, no jitter, no interrupts
- On-chip PWM generation for H-bridge and dump-load control
- Two-wire directional PWM interface for external FET drivers
- Programmable Vref via single-pin PWM input
- Low-latency error filtering and accumulator-based control loops
- Built-in dump-load minimum-pulse enforcement
- Dead-time-safe PWM outputs for discrete gate drivers
- Compact 3 mm × 3 mm package for dense power modules
- Low-voltage, low-power digital architecture
- Clean integration path for isolated voltage and current sensing
- Stable operation across wide input and load conditions
- Simple external component count — no MCU required
- Fully deterministic startup and shutdown behavior
- Compatible with micro-inverter assist and inverter-to-inverter sync
- Ideal for off-grid, backup, and energy-harvesting applications
- Designed for robust operation in noisy power environments

Typical Applications

- **Micro-grid formation and stabilization** for inverter-assist and off-grid backup systems
- **Energy-balancing controllers** using dump-load regulation for solar, storage, and hybrid sources
- **Compact power-processing modules** requiring precise AC synthesis and fast DC-link control

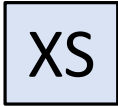
Typical Application



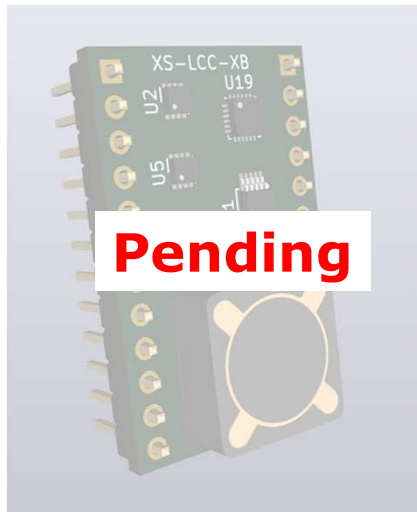
Description

The XS-PCC-01 sits at the center of a compact, low-voltage power-control system designed to synthesize a clean AC waveform while actively regulating the DC-link that feeds it. Differential isolated ADC inputs provide real-time sensing of both the AC output and the DC-link voltage, enabling the chip's dual-loop architecture to maintain stable operation across rapidly changing load and source conditions. An external H-bridge module, driven by the PCC-01's deterministic PWM outputs, generates the final AC waveform, while a dump-load MOSFET path absorbs excess energy to keep the DC-link within safe limits.

Around the PCC-01, the system integrates a small VFD-style rectifier and DC-link capacitor bank, isolated voltage and current sensors, and simple configuration elements such as potentiometers or strap pins for loop tuning. This architecture forms a complete micro-grid control core: the AC loop shapes the sine wave, the DC loop balances energy flow, and the dump-load path ensures stability during surplus generation. The result is a tightly integrated, minimal-component power-control platform suitable for experimentation, inverter-assist development, and compact energy-processing modules.



XS-PCC-01-XB Engineering Board (3D Render)



Typical Applications

- **Micro-grid experimentation** using low-voltage AC synthesis and DC-link control
- **Solar-inverter behavior studies** without high-voltage hardware
- **Rapid prototyping** of dump-load regulation, inverter-assist, and energy-balancing concepts

Purpose

A compact evaluation and experimentation board for engineers and solar enthusiasts to explore, prototype, and validate the XS-PCC-01 power-control architecture in a safe, low-voltage environment.

Features

- Integrates the XS-PCC-01 power-control ASIC
- Fully low-voltage design — safe for bench-top experimentation
- Differential ADC front-end brought out for external isolated sensors
- Two-wire directional PWM outputs for H-bridge driver evaluation
- Dedicated dump-load PWM output with minimum-pulse enforcement
- On-board Vref PWM generator with trimpot for loop tuning
- Breakout headers for all primary control and sense signals
- Independent AC and DC loop visibility for debugging and tuning
- Clean separation of analog sensing and digital control domains
- Compact board layout optimized for signal integrity
- Test points for AC error, DC error, accumulators, and dump-command paths
- Compatible with discrete MOSFET drivers and small H-bridge modules
- Supports rapid iteration of loop-gain and configuration strap settings
- Designed for stable operation with small bench-top power supplies
- Ideal for validating micro-inverter startup and shutdown behavior
- Provides a safe platform for exploring energy-balance algorithms
- No firmware required — deterministic hardware control loops
- Easy integration with oscilloscopes and logic analyzers
- Suitable for educational demonstrations of AC/DC control theory
- 3 mm XS-PCC-01 package showcased in a practical reference design